

Smart India Hackathon (SIH 2026) – Grand Finale Pitch

PROJECT: KRUSHIRAKSHA (■■■■■■■■■■) | PROBLEM ID: 26131 | GOVT OF MAHARASHTRA

Easy Hinglish Master Script (Aasani Se Samajhne Aur Samjhane Ke Liye) · Team: Binary Beacons

Speaker & Role	Kaam / Module (Hindi/English)	Time
Speaker 1: Problem Pitch Master (Team Lead)	Kisaan ki problem, Ramesh Patil ki story, 1.4 Lakh Crore ka loss aur KrushiRaksha ka intro	0:00 – 4:00 (4 min)
Speaker 2: ML Lead 1 (Vision & Live Camera)	Live YOLO Camera (38 FPS), Drone Video Stream, Photo Upload, Softmax Probabilities & Confusion Matrix	4:00 – 7:30 (3.5 min)
Speaker 3: ML Lead 2 (Weather & Epidemiology)	Live Weather Telemetry, Blitecast Fungal prediction, Safe Spray Window (Dawa chhidakne ka sahi time)	7:30 – 10:45 (3.25 min)
Speaker 4: ML Lead 3 (Pest Traps & CIBRC Dawa)	Pheromone Trap counting (Pink Bollworm), CIBRC certified chemical/organic dosage aur KVK alert	10:45 – 13:45 (3 min)
Speaker 5: Backend Engineer	FastAPI Backend, Outbreak GIS Heatmap (DBSCAN 10km Quarantine), Offline 2G SMS / IVR Toll-Free	13:45 – 16:45 (3 min)
Speaker 6: Frontend Developer	React 19 Fast UI, 11 Indian Languages translation, Audio Voiceout, Farmer/Officer/Govt portals	16:45 – 19:15 (2.5 min)
Speaker 1: Closing & Wrap-Up	Summary, pilot impact (4 days to 14ms latency), Govt integration aur Q&A; invitation	19:15 – 20:00 (45 sec)

PART 1: PROBLEM STATEMENT KO DUMDAAR TAREEKE SE PITCH KRNA (4 MINUTES)

Speaker 1: Team Lead & Problem Pitch Master | Time: 0:00 – 4:00

1. Story Se Start Karein (0:00 - 1:15):

"Respected Judges and Mentors, Good morning.

Hum hain **Team Binary Beacons**, aur hum present kar rahe hain hamara solution **Problem Statement ID 26131 — KrushiRaksha**.

Judges, sochiye Maharashtra ke Nashik jile ke ek kisaan Ramesh Patil ke baare mein. Ramesh ne apni saari jama-poonji—lagbhag 1.5 Lakh रुपये—apne 3 acre tamatar aur angoor ke khet me laga di. Fasal katne se do hafte pehle patto par chote-chote paani jaise kaale dhabbe dikhte hain.

Ramesh ke paas koi agricultural scientist nahi baitha hai. Desh ke 80% kisaano ki tarah wo ek patta tod kar dukaandar ke paas daudte hain. Dukaandar unhe bina soche-samjhe 2500 रुपये ki mehngi dawai thama deta hai. Ramesh aakar use khet me chhidak dete hain. Lekin 4 ghante baad tez baarish hoti hai, aur saari dawai beh jaati hai.

Teen din baad, wo fungus Late Blight poore khet me fail jaata hai. Poori 3 acre fasal sadh jaati hai. Ramesh ka 100% loss hota hai aur wo karz me doob jaate hain."

2. Bharat Ka Real Data Aur Problem (1:15 - 2:30):

"Judges, ye sirf Ramesh ki nahi, pure Bharat ke kisaano ki kahani hai:

- **1.4 Lakh Crore Ka Annual Loss:** ICAR ke mutabik, har saal Bharat me 15% se 25% fasal sirf bimariyo aur keedo ki wajah se barbaad ho jaati hai.

- **4 Din Ka Delay:** Jab tak kisaan ko bimari ka pata chalta hai aur lab report aati hai, tab tak 4 din nikal chuke hote hain aur tab tak bimari haath se nikal jaati hai.

- **Dawai Ka Galat Chhidkaav:** Sahi jaankari na hone par kisaan 35% zyada chemical daal dete hain, jisse mitti kharab hoti hai aur export reject ho jaata hai.

- **Officers Ki Kami:** Bharat me 1100 se zyada kisaan par sirf 1 extension officer hai. Har khet me daily jana impossible hai."

3. Purane Apps Kyu Fail Hue Aur KrushiRaksha Kya Hai (2:30 - 4:00):

"Play Store par hazaro apps hain, fir bhi ye solve kyu nahi hua? Kyunki:

1. Wo sirf ek photo mangte hain, real-time live camera ya drone scan nahi karte.

2. Wo mausam ko nahi dekhte—fungus hamesha tabhi badhta hai jab hawa me nami aur baarish ho.

3. Wo chemical companies ke agent ban kar sirf dawai bechte hain, sahi dosage aur organic tareeqa nahi batate.

Iska solution hai **KrushiRaksha**—ek complete AI platform jo Kisaan, Agricultural Officers aur Sarkar ko aapas me jodta hai. Aaiye hamare ML Engineer se dekhte hain iska Live AI Camera aur Model."

PART 2: LIVE AI CAMERA, MULTI-CLASS PROBABILITIES & CONFUSION MATRIX (3.5 MIN)

Speaker 2: ML Lead 1 (Vision & Edge AI) | Time: 4:00 – 7:30

Live Demo & Explanation:

"Thank you. Jaise aap screen par dekh sakte hain, Pillar 1 hai hamara **AI Diagnosis & Live YOLO Camera**:

- **YOLOv8 Real-Time Live Vision:** Kisaan ko 50 photo khinchne ki zarurat nahi hai. Wo bas live camera on karega. Hamara model **38 Frames Per Second** par chalta hai aur 14 millisecond me patto ke upar live bounding box bana kar bimari detect karta hai. Ye mobile camera, tractor camera aur drone video stream teeno ko support karta hai.
 - **Photo Upload & AI Vision API:** Jab kisaan photo upload karta hai, hamara model (Google Gemini 1.5 Flash Vision / ResNet-18) uske pixels scan karta hai aur Grad-CAM heatmap dikhata hai ki patte ke kis hisse me infection hai.
 - **Prediction Probabilities (Softmax Bars):** Hum model se sirf ek naam nahi nikaalte, balki har possible class ka probability score dikhate hain—jaise Mango Anthracnose: 87.4%, Healthy: 7.8%, Powdery Mildew: 3.1%. Isse clear pata chalta hai ki model ne pura analysis kiya hai.
 - **Confusion Matrix:** Upar diye Confusion Matrix button par click karke hum 54,000+ benchmark images par model ki **96.8% Accuracy** aur **0.962 F1-score** ko verify kar sakte hain.
- Lekin sirf photo se bimari nahi rukti, mausam sabse bada reason hota hai. Ab hamare 2nd ML Engineer mausam ke baare me batayenge."

PART 3: LIVE MAUSAM & SAFE SPRAY WINDOW (3.25 MINUTES)

Speaker 3: ML Lead 2 (Epidemiology & Weather AI) | Time: 7:30 – 10:45

Live Demo & Explanation:

"Thank you. Pillar 2 hai hamara **Live Weather Risk Forecaster**:

- **Live Weather Telemetry:** Humne Open-Meteo API se live agrometeorological data connect kiya hai. Kisaan 1-Click GPS dabayega aur uske khet ka live Temperature, Nami (Humidity), Patte ka geelapan (Leaf Wetness) aur Hawa ki speed screen par aa jayegi.
 - **Bimari Aane Se Pehle Prediction (Blitecast & 3-10 Rule):** Agar hawa me nami 85% se zyada hai aur patta 10 ghante geela rehta hai, to hamara algorithm kisaan ko **48 se 72 ghante pehle hi alert** kar deta hai ki fungus aane wala hai.
 - **Safe Spray Window (Dawai Chhidakne Ka Sahi Time):** Kisaan ka sabse bada loss tab hota hai jab wo dawai chhidakta hai aur baarish aa jaati hai. Hamara system 6 ghante ka rainfall probability aur wind speed check karke screen par bada-bada batata hai: **'SAFE SPRAY WINDOW ACTIVE'** ya **'UNSAFE SPRAY WINDOW'** taaki dawai kharab na ho.
- Ab hamare 3rd ML Engineer batayenge ki bimari aane par dawai ki sahi matra kaise nikaalte hain."

PART 4: PEST TRAP COUNTING & CIBRC APPROVED DAWAI DOSAGE (3.0 MINUTES)

Speaker 4: ML Lead 3 (Pest Traps & CIBRC Dosage) | Time: 10:45 – 13:45

Live Demo & Explanation:

"Thank you. Pillar 3 hai hamara **CIBRC Prescribed Dosage & IPM Calculator**:

- **3-Step Scientific Solution:** Bimari aane par hum 3 steps dete hain:
 1. Step 1 (Organic & Kheti Tareeqa): Kharaab patto ko khet se bahar jalana aur row-spacing theek karna.
 2. Step 2 (Biological Treatment): Trichoderma harzianum ya Pseudomonas jaise organic bio-agents.
 3. Step 3 (CIBRC Approved Chemical): Sarkar dwara registered chemical molecule (jaise Copper Oxychloride 50% WP ya Mancozeb) bina kisi company bias ke.
 - **Knapsack Tank Dosage Math:** Kisaan bas apna acre aur 15 Liter tank dalege, calculator exact gram dawai nikaal dega aur **PHI (Pre-Harvest Interval)** batayega ki kitne din tak fasal nahi todni hai taaki zahar na rahe.
 - **Pest Trap Counter (Pink Bollworm):** Kapas ke khet me lagaye gaye Pheromone trap ko hamara IP102 AI model count karta hai. Agar ek raat me **8 se zyada keede** pakde jaate hain, to system turant KVK ke agricultural scientist ko alert bhej deta hai.
- Ab hamare Backend Engineer batayenge ki ye pura system server aur offline me kaise chalta hai."

PART 5: BACKEND, OUTBREAK GIS & 2G OFFLINE SMS (3.0 MINUTES)

Speaker 5: Backend Engineer | Time: 13:45 – 16:45

Live Demo & Explanation:

"Thank you. Pillar 4 aur 5 hamara backend architecture hai:

- **FastAPI Backend:** Hamara backend Python FastAPI me bana hai jo multi-part video streams aur spatial queries ko 50ms se kam time me process karta hai.
 - **DBSCAN Outbreak GIS Map:** Jab alag-alag kisaan bimari scan karte hain, hamara spatial algorithm unke GPS ko cluster karta hai. Agar ek 10km ke area me case badhne lagte hain, to government dashboard par red zone aur **10km Quarantine Containment Zone** ban jaata hai taaki bimari pure state me na faile.
 - **Offline 2G SMS / IVR Gateway (*99*26131# & 1800-KRUSHI):** Agar kisaan ke paas smartphone ya 4G internet nahi hai, to wo sadharan keypad phone se SMS bhej sakta hai: `KRUSHI TOMATO SPOTS NASHIK`. Hamara backend 2 second me SMS ke zariye dawai ki sahi matra bhej deta hai bina kisi internet ke.
- Ab hamare Frontend Developer iske UI, 11 bhashao aur 3 portals ke baare me batayenge."

PART 6: FRONTEND, 11 INDIAN LANGUAGES & 3 ROLES (2.5 MINUTES)

Speaker 6: Frontend Developer | Time: 16:45 – 19:15

Live Demo & Explanation:

"Thank you. Kisaan ke liye technology tabhi useful hai jab wo uski bhasha me ho:

- **11 Indian Languages Support:** Screen par dekhiye, jaise hi hum Marathi, Punjabi, Kannada, Telugu, Tamil, Bengali, Gujarati, Malayalam, Odia ya Hindi select karte hain—pura website, buttons, camera controls, mausam aur dawai sab kuch turant translate ho jaata hai.
 - **Bol Kar Batane Wala Voiceout (Text-To-Speech):** 'Listen Advisory' dabane par app kisaan ko uski local bhasha me bol kar advice sunata hai.
 - **3 Dedicated Role Hubs:**
 1. ■ **Farmer Hub:** Simple mobile view for scan, weather & dosage.
 2. ■■■ **Officer Hub:** Kisaano ke khet inspection aur ticket solve karne ke liye.
 3. ■■ **Govt Command Center:** Pura state map, emergency dawa supply aur 50,000 kisaano ko ek saath alert bhejne ke liye.
- Ab main hamare Team Lead ko final wrap-up ke liye invite karta hoon."

PART 7: GRAND SUMMARY & FINAL WRAP-UP (45 SECONDS)

Speaker 1: Team Lead | Time: 19:15 – 20:00

"Judges, to summarize:

KrushiRaksha sirf ek concept ya prototype nahi hai—ye **100% working, coded aur live deployed** hai Vercel (krushiraksha.vercel.app) aur GitHub Pages par.

Hamari simulation me bimari pata chalne ka time **4 din se ghat kar 14 milliseconds** ho gaya, aur kisaan ke dawai ka kharcha **28% kam** hua.

Hum Maharashtra Sarkar ke MahaAgri portal ke saath integrate hone ke liye fully ready hain.

Thank you so much Judges! Ab hum aapke sawaalo ke jawab dene ke liye taiyaar hain."

JUDGES KE TOUGH SAWAAL & SIMPLE ANSWERS (CHEAT SHEET)

#	Judge Ka Sawaal	Kaun Bolega	Kya Bolna Hai (Simple Answer)
Q1	Aapka model baaki open-source models se alag kaise hai?	ML Lead 1 (Speaker 2)	Sir, standard models sirf clean white background par kaam karte hain. Humne PlantVillage aur IP102 dataset par dhoop, chhaya, camera shake aur drone video par train kiya hai, aur saath me Grad-CAM heatmap aur live weather context lagaya hai.

Q2	Agar kisaan anpadh hai ya uske paas smartphone nahi hai to?	Backend Lead (Speaker 5)	Sir, humne do solution banaye hain: 1. Button phone ke liye 2G SMS / IVR toll-free number (*99*26131#). 2. App me 11 bhashao ka Voiceout jo kisaan ko bol kar dawai sunata hai.
Q3	Pesticide company aapko paise dekar apni dawai promote karwa sakti hai?	ML Lead 3 (Speaker 4)	Sir, KrushiRaksha company ka naam nahi balki CIBRC certified chemical molecule (jaise Copper Oxychloride 50%) recommend karta hai. Brand names sirf reference ke liye hote hain, koi sponsored ads nahi hain.
Q4	Aapka business model kya hai?	Team Lead (Speaker 1)	Sir: 1. State Agriculture Departments ke liye B2G Outbreak GIS SaaS. 2. Drone spray companies ke liye API licensing. 3. Kisaano ke liye ye hamesha 100% Free rahega.